

Facade Glazing & 905 nm LiDAR Modeling for GCC High-Rise Commercial Buildings

TL;DR

- **GCC high-rise facades are overwhelmingly unitized curtain wall with solar-control/low-E coated insulated glass units (IGUs), which makes them PARTIAL MIRRORS at 905 nm — not voids.** Unlike clear glass ($\approx 4\%$ Fresnel reflectance per surface, mostly a LiDAR void), the silver-based solar-control and low-E coatings forced by the extreme-heat climate reflect roughly 20–76%+ of near-infrared, producing strong specular returns and mirror/ghost artifacts near normal incidence.
 - **The most reliable hard returns are the aluminium mullion/transom grid**, a near-periodic lattice with $\sim 50\text{--}65$ mm face widths on a $\sim 1.2\text{--}1.8$ m horizontal module and a $\sim 3.5\text{--}4.5$ m floor-to-floor vertical module; this grid (plus opaque spandrel zones at every floor line) should be the backbone of corner/edge detection, while the glass infill returns are angle-dependent and unreliable.
 - **The single biggest empirical gap: no public glass datasheet reports a discrete 905 nm value** — all are band-integrated (300–2500 nm per ISO 9050/EN 410). You must extract 905 nm from full spectral curves (LBNL International Glazing Database) or measure your actual panels with a UV-VIS-NIR spectrophotometer; treat all numbers here as priors to tune against your own scans.
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Key Findings

1. **Dominant system:** Modern GCC commercial towers use unitized (factory-glazed, one-floor-tall) aluminium curtain wall almost universally for high-rise speed and quality. Stick-built systems appear on low/mid-rise and podiums.
2. **Glass is coated, not clear:** Because cooling loads dominate Gulf design, building codes and economics push nearly all vision glass to solar-control and/or low-E coated IGUs. As a concrete anchor, **Dubai's Al Sa'fat Green Building system (2nd ed., Jan 2023) mandates glazing with a Solar Heat Gain Coefficient (SHGC) ≤ 0.25** for its Silver baseline — a value effectively unachievable without a solar-control/low-E coating. Saudi SBC 601/602 and Abu Dhabi Estidama impose comparable fenestration SHGC/U-factor limits. Clear uncoated vision glass is therefore rare on a conditioned commercial facade. (Daralnaseeb)
3. **Optical split at 905 nm:** Clear and body-tinted glass behave as near-voids (low reflectance $\sim 4\text{--}8\%$); coated solar-control and low-E glass behave as partial-to-strong mirrors (NIR reflectance $\sim 20\text{--}76\%$).

4. **Geometry is highly regular and randomizable:** Panel sizes, mullion widths, grid modules, and spandrel/vision alternation fall in tight, well-documented ranges suitable for parametric scene generation.
 5. **IGUs and laminates create multi-surface returns:** Two or three parallel glass panes separated by 12–16 mm gaps, plus ~0.76 mm PVB interlayers ($n \approx 1.48$), produce closely spaced multiple/ghost returns at known small separations.
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PART A — Facade Glazing Systems & Geometry

A1. Curtain wall systems: stick vs unitized

- **Unitized** dominates high-rise: factory-assembled units, typically **one floor tall** × **one module wide**. Module widths most commonly **1.2–1.5 m**, with the broader practical range **1.2–1.8 m** (occasionally to 2.4 m for shopfront/podium). Unit height = floor-to-floor.
- **Stick-built:** site-assembled mullions/transoms then glazed; used on lower-rise and complex geometry. Vision panel widths commonly **0.6–1.5 m**.
- **Vision panel dimensions (typical):** width **1.2–1.5 m**, height **1.5–3.0 m** (vision portion of a floor). Maximum fabricable glass lights reach ~**3.3 m** × **6.0 m** but are not the norm.
- **Floor-to-floor module:** commercial offices typically **3.6–4.5 m**; a common design value is ~**4.0 m**. Tiling: units stack vertically one-per-floor and repeat horizontally on the module width, producing a near-perfect periodic grid.

A2. Mullion/transom grids

- **Face (sightline) width:** commonly **50–65 mm** (typical captured systems 50 mm; range 25–60 mm for slim systems, up to 100 mm for expressed mullions). Example regional/standard sizes: 50 mm and 60 mm sightline systems (Aluprof MB-SR50N/60N), 65 mm × 125 mm unitized profiles (Gutmann UCW-65).
- **Frame depth (into building):** **150–225 mm** for high-rise unitized (up to 250 mm in supertall/high-wind); stick systems often 50–125 mm depth depending on span.
- **Protrusion from glass plane:** Captured systems have pressure plates/caps that stand proud of the glass by roughly **10–30 mm**; structurally-glazed (SSG) systems are nearly flush. Expressed "fin" mullions can protrude much more (decorative).
- **Grid spacing:** horizontal **1.2–1.8 m** (module), vertical = floor-to-floor **3.6–4.5 m**, often subdivided by transoms into a vision band + spandrel band. Some facades use a constant module such as 1.5 m.
- **Joints between unitized panels:** vertical (split mullion) **10–15 mm**; horizontal stack joint **15–25 mm** (to 35 mm seismic). These are silicone-sealed gaps.

A3. Structural / point-fixed (spider) & silicone-bonded (SSG) glazing

- **SSG (structural sealant glazing):** glass bonded to frame with structural silicone; the external face shows only a **silicone joint** between lights, typically **flush, ~12–20 mm wide**. To LiDAR there is effectively **no protruding frame** — just a thin recessed/flush seam. Example: flush SSG systems advertise a **21 mm external sightline** that is itself a silicone joint (AMS MU800 SG). (Ams)
- **Point-fixed (spider) glazing:** stainless spider fittings clamp glass corners. Spider arm spans roughly **150–300 mm** (Dimensions.com cites fitting widths of 15/20/30 cm), fitting depth **50–100 mm**, mounted on a structural backup (mullion / glass fin / cable truss). Glass is typically 10–12 mm toughened. The LiDAR "frame" signal here is the **discrete metal spider bodies + bolts** (small bright specular points) rather than a continuous grid — much sparser hard-return geometry.
- Implication: on SSG/spider facades the reliable periodic-grid prior weakens; the dominant returns are the coated glass itself plus sparse metal fittings.

A4. Spandrel vs vision glass

- **Arrangement:** Spandrel (opaque) panels cover floor slabs, perimeter beams and ceiling/MEP plenum zones at each floor line; vision glass covers the eye-level band. They alternate vertically every floor, producing strong horizontal banding at the floor-to-floor pitch.
- **Spandrel construction:** ceramic-fritted or opacified/shadow-box glass, often the same coated outer lite as vision glass for visual continuity, backed by insulation.
- **Vision-to-spandrel ratio:** governed by Window-/Wall-Ratio limits. Dubai Municipality's Green Building Regulations classify glazing below 40% of built-up area as the "<40%" tier and additionally require that at least 50% of the total glazed surface area be oriented away from peak south/west solar exposure; regional low-E products are explicitly marketed for "facades with 40–60% glazed area" (Emirates EmiCool Supervision 40-21). A typical tower therefore runs roughly **40–60% vision, 40–60% spandrel** by facade area, varying by orientation and code path. (Wkcgrouop)

A5. Frame materials & NIR behavior

- **Material:** Extruded aluminium (6063/6061), finished either **anodized** or **polyester powder-coated** (Qualicoat).
- **NIR behavior:** Bare/anodized aluminium is **highly reflective in NIR** — published measurements show even black-dyed anodizing is "quite reflective at red and near-IR wavelengths (>700 nm)" (arXiv 1407.8265, black-materials reflectivity study). Powder-coat color drives NIR: light colors reflect strongly; dark/black powder coats absorb visible but NIR depends on pigment. **Net: metal frames generally give strong, often near-specular**

returns at 905 nm — good for grid detection, but specularity means returns are angle-dependent (bright near-normal, dim at grazing).

Summary table — geometric ranges for randomized scene generation

Parameter	Typical / common	Full range	Notes
Vision panel width	1.2–1.5 m	0.6–1.8 m (to 2.4 m podium)	= unitized module width
Vision panel height	1.5–3.0 m	up to ~3.6 m	vision band of a floor
Max fabricable light	—	~3.3 × 6.0 m	rarely used
Floor-to-floor (module H)	~3.6–4.0 m	3.5–4.5 m	vertical unit pitch
Mullion/transom face width	50–65 mm	25–100 mm	slim SSG ~21 mm seam
Frame depth (into bldg)	150–225 mm	50–250 mm	stick shallower
Frame protrusion from glass	10–30 mm	0 (SSG flush) - >50 mm (fins)	
Horizontal grid module	1.2–1.8 m	—	often constant 1.5 m
Vertical (split) joint	10–15 mm	—	silicone
Horizontal stack joint	15–25 mm	to 35 mm seismic	silicone
IGU cavity (air/argon)	12–16 mm	6–20 mm	parallel-surface separation
DGU total thickness	26–32 mm	to 42 mm laminated	
PVB interlayer	0.76 mm	0.38–1.52 mm	n≈1.48
Spider arm span	150–300 mm	—	point-fixed only
Vision : spandrel	~50:50	40–60% either	per WWR/code

PART B — Glass Types & 905 nm Optical Properties

Critical framing finding: No public architectural glass datasheet reports a discrete value at 905 nm. All manufacturer figures and governing standards (ISO 9050:2003, EN 410, JIS R 3106, NFRC 300) are band-integrated — visible (380–780 nm) and solar-direct (300–2500 nm). **905 nm values must be interpolated from full 300–2500 nm spectral curves (LBNL International Glazing Database, ~5 nm resolution) or measured empirically.** This is the key gap to close with your own spectrophotometry.

B1-B2. Type-by-type (reflectance R / transmittance T / absorptance A near 905 nm)

- **Clear float glass:** $R \approx 4\%$ **per air-glass interface** (Fresnel, $n \approx 1.5$; ScienceDirect "Float Glass" overview states "about 4%... in the $0.4 < \lambda < 2\text{-}\mu\text{m}$ range, implying maximum transmittance for a glass pane is 92%"), **~7–8% total two-surface**. Near 900 nm T is still high (**~82–87%**), just before the **iron-oxide ($\text{Fe}_2\text{O}_3/\text{FeO}$) absorption dip centered ~1000 nm**. → **LiDAR VOID** (weak specular return only near normal incidence; most NIR transmitted). (ScienceDirect)
- **Body-tinted (bronze/grey/green/blue):** Attenuates by **absorption, not reflection**. NIR reflectance stays **~4–8% ($\approx$ clear)** while NIR transmittance drops sharply (tinted IR transmittance **~22–51%** for 4 mm, per PPG patent data). Spectrophotometric measurement (Shimadzu UV-VIS-NIR notes) finds transparent vs tinted glasses differ greatly in transmittance but have "about the same" reflectance in the infrared. → **Still largely a LiDAR VOID for returns** (low reflectance); it just transmits less. Tinted $\neq$ reflective for LiDAR. (Shimadzu)
- **Solar-control / reflective coated (metallic pyrolytic hard-coat or magnetron-sputtered silver):** Engineered to reflect solar NIR. External reflectance **~20–50%+**. A peer-reviewed sol-gel multilayer anchors the in-band value: Nagamedianova et al., "Solar heat reflective glass by sol-gel nanostructured multilayer coatings," *Optical Materials* 2011, Vol. 33, pp. 1999–2005, a $\text{TiO}_2/\text{SiO}_2/\text{TiO}_2$ dip-coated stack with "high visible transmittance $>70\%$ combined with high solar reflectance about 30% (with **reflectivity up to 60% at region from 800 to 950 nm**) and high UV blockage." Regional product anchor: **Emirates Float Glass LLC EmiCool DSR** states it "provides high external reflectance with reflectance values reaching **up to 40%**" with light transmission 20–40% (integrated value, not a 905 nm point). → **PARTIAL-TO-STRONG MIRROR at 905 nm**. (Efgme)
- **Low-E coated (silver-based soft-coat, single/double/triple silver):** Designed to block NIR. The strongest in-band anchor: "**Improvement of optical properties and thermal stability in the $\text{TiO}_2/\text{Si}_3\text{N}_4/\text{Ag}/\text{Si}_3\text{N}_4/\text{TiO}_2$ low-emissivity coating**" (ScienceDirect S2468606925000838) reports "the T_{vis} reaches 0.85, the **RNIR reaches 0.76**, and both the T_{max} and R_{max} reach 0.92" (magnetron-sputtered). Regional anchor: **EmiCool Plus T (MSVD sputtered Low-E)** datasheet specifies **~63–65%** light transmission and is "highly effective in blocking the transmission of heat in the near and far infrared (IR) wavebands." → **STRONGEST 905 nm reflector among facade glasses**. Note triple-silver low-E transmits

near-zero in 1000–2500 nm; the 800–1000 nm region is the steep transition where reflectance climbs. (Emiratesglass)

- **Ceramic-fritted (spandrel):** Opaque enamel fused to glass. Peer-reviewed measurement (Challenging Glass proceedings) shows that when an NIR-reflective solar-control coating is present, the fritted pane's NIR reflectance is "significantly higher" than without it; frit itself is a diffuse/absorbing opaque layer. → **Opaque return (not a void); behaves like a diffuse/partly specular solid surface.** (Challengingglass)
- **Laminated (PVB):** PVB $n \approx 1.48$, ~0.76 mm typical. Adds two extra low-index-contrast interfaces (small Fresnel mismatch glass↔PVB), so weak extra reflections; PVB transmits NIR well. → **Minor extra weak returns; bulk behavior follows the glass/coating.**
- **IGU (double/triple glazed):** Two or three panes + 12–16 mm gaps. Each glass surface is a Fresnel interface; the coated surface dominates. → **Multiple parallel returns at known mm-scale separations.**

B3. Realistic prior: how likely is a panel coated/reflective?

Given Gulf cooling loads and code mandates (Dubai Al Sa'fat SHGC ≤ 0.25 , Abu Dhabi Estidama, Saudi SBC 601/602 fenestration SHGC limits), **the realistic prior is that the great majority of vision panels on a modern conditioned GCC commercial tower carry a solar-control and/or low-E coating** — clear uncoated vision glass is rare. A reasonable simulator prior: **~85–95% of vision panels coated (partial mirror), ~5–15% clear/lightly-tinted (void-like)**, with spandrel panels (40–60% of facade) opaque. Treat the exact split as tunable; it is a defensible engineering estimate, not a published population statistic.

B4. Multi-interface / ghosting from IGUs & laminates

- A double IGU presents four weak parallel reflectors (outer-pane front/back, inner-pane front/back) separated by glass thickness (~6 mm) and the **air/argon gap (12–16 mm)**. A laminate adds a **~0.76 mm PVB** sublayer.
 - For a single-return 905 nm LiDAR these typically merge (separations 6–20 mm are below most range resolution), but a sensitive/multi-echo sensor can register **closely spaced multiple returns / range walk / ghost points** at these fixed small offsets. On strongly reflective coated IGUs, specular reflection can also produce **mirrored geometry** (false copies of nearby objects or the drone itself) — a documented LiDAR failure mode ("ghost points" from multi-path glass returns).
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PART C — Implications for 905 nm LiDAR

C1. Detectable returns vs voids

- **Coated vision glass (the GCC norm): mostly DETECTABLE** as elevated-intensity specular returns near normal incidence, fading rapidly off-normal. Because most panels are coated, a GCC facade is **more LiDAR-visible than a generic clear-glass building** — but in an angle-dependent, specular way. Note that LiDAR intensity peaks on glass are only reliably detectable where the beam is near-perpendicular; off-normal, returns collapse (consistent with the glass-detection literature, e.g., MDPI Sensors 21:2263).
- **Clear/tinted glass (minority): VOIDS** — frequently missed; the beam transmits and returns the interior/background or nothing. (nih)
- **Spandrel + frames: reliably DETECTABLE** opaque/metal returns.

C2. Ghost / mirror artifacts

- Reflective coatings make **specular ghosting a primary failure mode**: mirrored false points, duplicated geometry, and (on IGUs/laminates) closely spaced multi-returns. Prevalence scales with the coated-panel fraction (high in GCC) and with near-normal incidence geometry. A facade-cleaning drone typically hovers facing the facade — the worst case for specular glare and ghost generation.

C3. Reliable geometry for corner/edge detection

- **The mullion/transom grid is the dependable skeleton**: near-periodic, ~50–65 mm members on a 1.2–1.8 m × 3.6–4.5 m lattice, metal (strong NIR returns), with consistent floor-banding from spandrel zones. **Build detection/localization on the frame lattice and spandrel bands, not on glass returns.** Treat glass infill as a stochastic, angle-dependent surface; treat SSG/spider facades as the hard case (sparse hard returns: thin silicone seams + isolated spider fittings).

Recommendations

1. **Model the frame grid as ground-truth geometry.** Randomize: module width U(1.2, 1.8) m; floor-to-floor U(3.5, 4.5) m; mullion face width U(50, 65) mm (allow 21 mm SSG seam and up to 100 mm expressed); protrusion U(10, 30) mm captured / 0 for SSG; stack joint 15–25 mm; split joint 10–15 mm.
2. **Assign glass classes by prior:** ~85–95% coated (partial mirror), remainder clear/tinted (void); 40–60% of facade area spandrel (opaque). Tune these fractions to the specific city/era of building once you have scans.

3. **Model 905 nm reflectance per class:** clear/tinted $R \approx 4-8\%$ (void); solar-control $R \approx 20-50\%$ (e.g., sample U(0.20, 0.50), informed by the 60%-at-800-950 nm sol-gel ceiling and EmiCool DSR's $\sim 40\%$ integrated); low-E $R \approx 50-80\%$ (e.g., U(0.50, 0.80), anchored on the 0.76 RNIR coating); metal frame strong specular; spandrel diffuse-opaque. **Make all reflectance strongly angle-dependent** (specular lobe + small diffuse term); retroreflection only near normal incidence.
4. **Inject IGU/laminate multi-surface returns:** parallel reflectors at ~ 6 mm glass, 12-16 mm gap, 0.76 mm PVB; enable ghost/mirror points off coated panels and model range-walk where echoes merge below range resolution.
5. **Close the empirical gap first.** Before trusting the sim, measure 905 nm reflectance/transmittance vs incidence angle on representative GCC panels (UV-VIS-NIR spectrophotometer + integrating sphere per ISO 9050; ideally a 905 nm laser source at varying angle). Pull raw spectral curves from the LBNL IGDB for named products (Guardian SunGuard, Emirates EmiCool DSR/Plus T, Vitro Solarban, AGC, NSG/Pilkington, Şişecam) and read the 905 nm point directly.

Benchmarks that change the plan:

- If measured coated-panel NIR reflectance at operational angles is consistently **>30%**, prioritize ghost/mirror rejection (multi-echo discrimination, specular-outlier filtering).
- If **<10%** (mostly clear/tinted stock, or you operate at steep grazing angles), prioritize void-handling and frame-only SLAM.
- If target-market facades are **SSG/spider** (minimal frames), invest in glass-surface detection (intensity-variance / multi-echo methods) because the periodic-grid prior fails.

Caveats

- **No 905 nm-specific public glass data exists; every optical value here is interpolated from band-integrated specs or research coatings and MUST be validated by your own measurement.**
- Two of the strongest quantitative coating figures — 60% reflectivity at 800-950 nm (Nagamedianova et al. 2011) and 0.76 RNIR low-E (ScienceDirect S2468606925000838) — are from research-grade coatings, not facade product datasheets; they are directionally correct and in-band but not product-exact.
- Manufacturer "external reflectance up to 40%" (EmiCool DSR) is an integrated value, not a 905 nm point.
- Vision:spandrel ratio and the coated-panel prior are engineering estimates from code/WWR practice, not measured population statistics.
- Geometric ranges are global curtain-wall norms corroborated across multiple sources; GCC towers follow these but specific landmark towers may deviate (custom megapanel,

expressed fins, double-skin facades — e.g., Abu Dhabi double-skin residential research in MDPI Buildings 13:2148).

- Some sourcing leans on system-manufacturer literature (Aluprof, AMS, Gutmann, Emirates Float Glass) and aggregator/architectural references rather than primary metrology; cross-validate critical numbers against the IGDB and your own scans before committing simulator parameters.